

TE Computer (V) Sub: Theory of Computation Date: 30th Sept 2025

CO5	Develop and fundamental understanding of computation through Turing Machine, and acquire knowledge of the decidability and un-decidability.
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Assignment No 04

1. Topic: Turing Machines
2. Construct TM to check well-formed ness of parenthesis
3. Design TM for unary adders
4. Design TM which recognized word of the form $anbn$ where $n \geq 1$ (same for 0,1,2)
5. Design TM to accept all even palindromes over a,b.
6. Write sort note on variants of Turing Machine.
7. Write sort note on Applications, Power and Limitations of TMs.

TE Computer (V) Sub: Theory of Computation Date: 8th Oct 2025

CO5	Develop and fundamental understanding of computation through Turing Machine, and acquire knowledge of the decidability and un-decidability.
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Assignment No 05

Topic: Undecidability

1. Write a short note on
 - a. Recursive and recursively enumerable language
 - b. Halting problem
 - c. Rice's Theorem
 - d. Post Correspondence Problem